INTRODUCTION

In the last few chapters the process of data collection was discussed. The collected data is usually contained in schedules and questionnaires. But that is not in an easily assimilable form. The answers will require some analysis if their salient points are to be brought out. As a rule, the first step in the analysis is to classify and tabulate the information collected, or, if published statistics have been employed, rearrange these into new groups and tabulate the new arrangement. In case of some investigations, the classification and tabulation may give such a clear picture of the significance of the material arranged that no further analysis is required. In other cases, these processes, though may materially assist the analysis, are not sufficient presentation of the facts. They are, however, very important whether they complete the analysis or form only part of it. The questionnaire may have been very carefully drawn up and the answers may be both complete and accurate, but until these answers are all brought together into the class to which they belong and the whole information displayed in a tabular form, no one will be a great deal wiser as to the contents of the replies.

Although the phrase "classification and tabulation" has been used, classification is, in effect, only the first step in tabulation, for, in general, items having common characteristics must be brought together before the data can be displayed in tabular form.

MEANING AND OBJECTIVES OF CLASSIFICATION

After collection and editing of data the first step towards further processing the same is classification. Classification is the grouping of related facts into classes. Facts in one class differ from those of classification. Sorting facts on one basis of classification and then on another basis is called cross-classification. This process can be repeated as many times as there are possible bases of classification. Classification of data is a function very similar to that of sorting letters in a post-office. It is well known that letters collected in a post-office are sorted into different lots on a geographical basis, *i.e.*, in accordance with their destinations such as Mumbai, Calcutta, Kanpur, Jaipur, etc. They are then put into separate bags, each containing letters with a common characteristic, *viz.*, having the same destination. Classification of statistical data is comparable to the sorting operation. To take another example, when students seek admission in a college they submit applications to the office. The application forms contain particulars about their performance in the previous examinations, their date of birth, sex, nationality, etc. If one is interested in finding out how many first, second and third class students have joined the college, one may look into each and every form and note whether it relates to a first class student, second class student etc. He may find that out of 1,000 students who took admission 50 had first class, 800 second class and 150 third class. The process with the help of which this information in a summary form is obtained is called the classification of data.

Objects of Classification

The principal objectives of classifying the data are:

- To condense the mass of data in such a manner that similarities and dissimilarities can be readily apprehended. Millions of figures can thus be arranged in a few classes having common features.
- To facilitate comparison.
- To pinpoint the most significant features of the data at a glance.
- To give prominence to the important information gathered while dropping out the unnecessary elements.
- To enable a statistical treatment of the material collected.

TYPES OF CLASSIFICATION

Broadly, the data can be classified on the basis of following four criteria: criteria:

- Geographical, *i.e.*, area-wise, *e.g.*, cities, districts, etc.
- Chronological, *i.e.*, on the basis of time.
- Qualitative, *i.e.*, according to some attributes.
- Quantitative, *i.e.*, in terms of magnitudes.

Geographical Classification

In this type of classification data are classified on the basis of geographical or vocational differences between the various items, like States, cities, regions, zones, areas, etc. For instance, the production of foodgrains in India may be presented State-wise in the following manner:

STATE-WISE ESTIMATES OF PRODUCTION OF FOODGRAINS

Name of State	*Production of Foodgrains (Thousand tonnes)*
Andhra Pradesh	1093.7
Bihar	12899.0
Haryana	11334.7
Punjab	21148.9
Uttar Pradesh	41828.6
All India	1,92,433.6

Geographical classifications are usually listed in alphabetical order for easy reference. Items may also be listed by size to emphasise the important areas as in ranking the States by population. Normally, in reference table the first approach is followed and in summary tables the second approach is followed.

Source: Economic Survey; Government of India, 2003-04.

Chronological Classification

When data are observed over a period of time the type of classification is known as chronological classification. For example, we may present the figures of population (or production, sales, etc.) as follows:

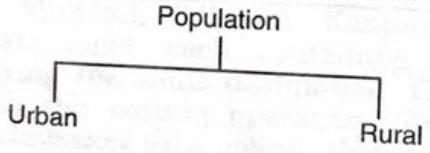
POPULATION OF INDIA FROM 1951 TO 2001

Year	*Population (in crores)*	*Year*	*Population (in crores)*
1951	36.11	1981	68.33
1961	43.92	1991	84.63
1971	54.82	2001	102.37

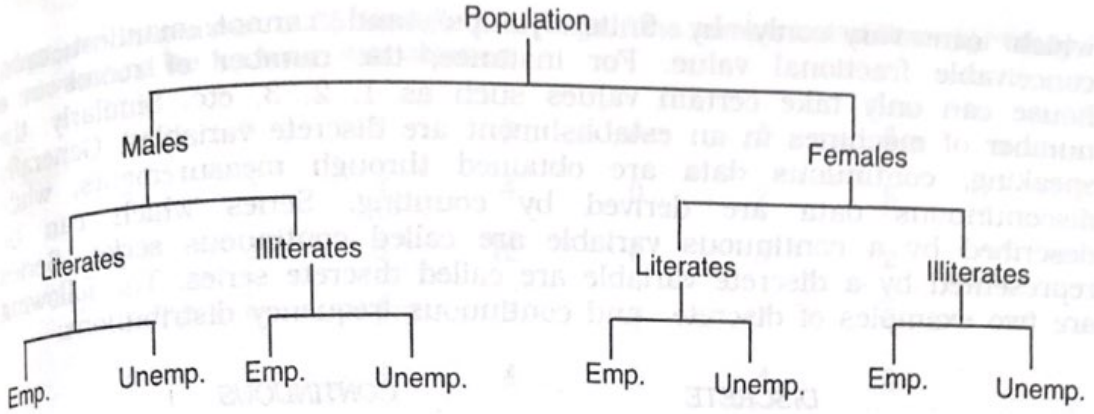
Time series are usually listed in chronological order, normally starting with the earliest period. When the major emphasis falls on the most recent events, a reverse time order may be used.

Qualitative Classification

In qualitative classification data are classified on the basis of some attribute or quality such as sex, colour of hair, literacy, religion, etc. The point to note in this type of classification is that the attribute under study cannot be measured: one can only find out whether it is present or absent in the units of the population under study. For example, if the attribute under study is population, one can find out how many persons are living in urban area and how many in rural area. Thus when only one attribute is studied two classes are formed, one possessing the attribute and the other not possessing the attribute. This type of classification is known as simple classification. For example, the population under study may be divided into two categories as follows:



In a similar manner, we may classify population on the basis of sex, *i.e.*, into males and females, or literacy, *i.e.*, into literate and illiterate, and so on. The type of classification where only two classes are formed is also called two-fold or dichotomous classification. If instead of forming only two classes we further divide the data on the basis of some attribute or attributes so as to form several classes, the classification is known as manifold classification. For example, we may first divide the population into males and females on the basis of the attribute 'sex' ; each of these classes may be further subdivided into 'literate' and 'illiterate' on the basis of the attribute 'literacy'. Further classification can be made on the basis of some other attribute, say, employment. An example of manifold classification is given here:



Note. Emp. indicates Employed and Unemp. indicates Unemployed.

Quantitative Classification

Quantitative classification refers to the classification of data according to some characteristics that can be measured, such as height, weight, income, sales, profits, production, etc. For example, the students of a college may be classified according to weight as follows:

Weight (in lb.)	*No. of Students*
90–100	50
100–110	200
110–120	260
120–130	360
130–140	90
140–150	40
Total	1,000

Such a distribution is known as empirical frequency distribution or simple frequency distribution.

In this type of classification, there are two elements, namely *(i)* the variable, *i.e.*, the weight in the above example, and *(ii)* the frequency, i.e., the number of students in each class. There were 50 students having weight ranging from 90 to 100 lb., 200 students having weight ranging from 100 to 110 lb., and so on. Thus we can find out the ways in which the frequencies are distributed.

A frequency distribution refers to data classified on the basis of some variable that can be measured such as prices, wages, age, number of units produced or consumed. The term 'variable' refers to the characteristic that varies in amount or magnitude in a frequency distribution. A variable may be either continuous or discrete. A continuous variable, also called continuous random variable, is capable of manifesting every conceivable fractional value within the range of possibilities, such as the height or weight of persons or the weight of a product. In a continuous variable, thus, data are obtained by numerical measurements rather than counting. For example, when a student grows, say, from 90 cm. to 150 cm., his height passes through all values between these limits. On the other hand, a discrete variable is that

which can vary only by finite "jumps" and cannot manifest ever conceivable fractional value. For instance, the number of rooms in house can only take certain values such as 1, 2, 3, etc. Similarly, th number of machines in an establishment are discrete variables. Generall speaking, continuous data are obtained through measurements, whil discontinuous data are derived by counting. Series which can b described by a continuous variable are called continuous series. Series represented by a discrete variable are called discrete series. The following are two examples of discrete and continuous frequency distributions:

DISCRETE		*CONTINUOUS*	
No. of Children	*No. of Families*	*Weight (lb.)*	*No. of Persons*
0	10	100–110	10
1	40	110–120	15
2	80	120–130	40
3	100	130–140	45
4	250	140–150	20
5	150	150–160	4
6	50		
Total	680	**Total**	134
(a) Discrete Frequency Distribution		*(b) Continuous Frequency Distribution*	

Although the theoretical distinction between continuous and discrete variations is clear and precise, in practical statistical work it is only an approximation. The reason is that even the most precise instruments of measurement can be used only to a finite number of places. Thus, every theoretically continuous series can never be expected to flow continuously with one measurement touching another without any break in actual observations.

FORMATION OF A DISCRETE FREQUENCY DISTRIBUTION

The process of preparing this type of distribution is very simple. We have just to count the number of times a particular value is repeated which is called the frequency of that class. In order to facilitate counting prepare a column of "tallies". In another column, place all possible values of variable from the lowest to the highest. Then put a bar (vertical line) opposite the particular value to which it relates. To facilitate counting, blocks of five bars are prepared and some space is left in between each block. We finally count the number of bars and get frequency.

The process shall be clear from the following examples:

Illustration 1. In a survey of 35 families in a village, the number of children per family was recorded and the following data obtained:

1	0	2	3	4	5	6
7	2	3	4	0	2	5
8	4	5	12	6	3	2
7	6	5	3	3	7	8
9	7	9	4	5	4	3

Represent the data in the form of a discrete frequency distribution.

Solution.

FREQUENCY DISTRIBUTION OF THE NUMBER OF CHILDREN*

No. of Children	*Tallies*	*Frequency*
0	||	2
1	|	1
2	||||	4
3	𝍸 |	6
4	𝍸	5
5	𝍸	5
6	|||	3
7	||||	4
8	||	2
9	||	2
10	—	0
11	—	0
12	|	1
		Total 35

It is clear from the table that the number of children varied from 0 to 12. There were 2 families with no child, 5 families with 4 children each and only one family with 12 children.

Illustration 2. Count the number of letters in each word of the para given below (ignoring comma, full-stop, etc.) and prepare a discrete frequency distribution.

"Today, to a very striking degree, our culture has become a statistical culture. Even a person who may never have heard of an index number, is attached in an intimate fashion by the gyrations of those index numbers which describe the cost of living."

—*Morris Hamburg*

* A frequency distribution or frequency table is simply a table in which the data grouped into classes and the number of cases which fall in each class are recorded. The numbers in each class are referred to as 'frequencies', hence the term 'frequency'. When the number of items are expressed by their proportion in each class, the table is usually referred to as a 'relative frequency distribution', or simply a 'percentage distribution.

TABULATION OF DATA

One of the simplest and most revealing devices for summarizing data and presenting them in a meaningful fashion is the statistical table*. A table is a systematic arrangement of statistical data in columns and rows. Rows are horizontal arrangements whereas columns are vertical ones. The purpose of a table is to simplify the presentation and to facilitate comparisons. The simplification results from the clear-cut and systematic arrangement, which enables the reader to quickly locate desired information. Comparison is facilitated bringing related items of information close together.

Difference between Classification and Tabulation

The chapter heading 'Classification and Tabulation of Data' should not lead the reader to believe that these are two distinct processes. In fact, they go together, classification being the first step in tabulation. Before the data are put in tabular form they have to be classified, *i.e.*, the different items having common characteristics must be brought together. It is only after this step that the data are displayed under different columns and rows so that their relationship can be easily understood.

Role of Tabulation

Tables make it possible for the analyst to present a huge mass of data in a detailed orderly manner within a minimum of space. Because of this, tabular presentation is the cornerstone of statistical reporting.

The significance of tabulation will be clear from the following points:

(i) It simplifies complex data When data are tabulated all unnecessary details and repetitions are avoided. Data are presented systematically in columns and rows. Hence, the reader gets a very clear idea of what the table presents. There is thus a considerable saving in time taken in understanding what is represented by the data and all confusion is avoided. Also a large amount of space is saved because of non-duplicating of his headings and designations; the description at the top of a column serves for all the terms beneath it.

(ii) It facilitates comparison Tabulation facilitates comparison. Since a table is divided into various parts and for each part there are totals and sub-totals, the relationship between different parts of data can be studied much more easily with the help of a table than without it.

(iii) It gives identity to the data When the data are arranged in a table with a title and number they can be distinctly identified and can be used as a source reference in the interpretation of a problem.

* "A statistical table is the logical listing of related quantitative data in vertical columns and horizontal rows of numbers with sufficient explanatory and qualifying words, phrases and statements in the form of titles, headings and notes to make clear the full meaning of data and their origin." —*Tuttle*
"A table summarizes the data by using columns and rows and entering figures in the body of the table. —*Karmel*

(iv) It reveals patterns Tabulation reveals patterns within the figure which cannot be seen in the narrative form. It also facilitates the summation of the figures if the reader desires to check the totals.

PARTS OF A TABLE

The number of parts of a table varies from case to case depending upon the given data. However, the main parts of a table in general are discussed here.

1. Table Number Each table should be numbered. There are different practices with regard to the place where this number is to be given. The number may be given either in the centre at the top above the title or inside of the title at the top or in the bottom of the table on the left-hand side. However if space permits the table number should be given in the, centre as is shown in the specimen table given on page. Where there are many columns, it is also desirable to number each column so that easy reference to it is possible.

2. Title of the Table Every table must be given suitable title. The title is a description of the contents of the table. A complete title has to answer the questions what, where and when in that sequence. In other words,

(a) what precisely are the data in the table (*i.e.*, what categories of statistical data are shown)?

(b) where the data occurred (*i.e.*, the precise geographical, political or physical area covered)?

(c) when the data occurred (*i.e.*, the specific time or period covered by the statistical materials in the table)?

The title should be clear, brief and self-explanatory. However, clarity should not be scarificed for the sake of brevity. Long titles cannot be read as promptly as short titles, but at times they may have to be used for the sake of clarity. The title should be so worded that it permits one and only one interpretation. It should be in the form of a series of phrases rather than complete sentences. Its lettering should be the most prominent of any lettering on the table.

3. Caption Caption refers to the column headings. It explains what the column represents. It may consist of one or more column headings. Under a column heading there may be sub-heads. The caption should be clearly defined and placed at the middle of the column. If the different columns are expressed in different units, the units should be mentioned with the captions. As compared with the main part of the table the caption should be shown in smaller letters. This helps in saving space.

4. Stub As distinguished from caption, stubs are the designations of the rows or row headings. They are at 'the extreme left and perform the same function for the horizontal rows of numbers in the table as the column headings do for the vertical columns of numbers. The stubs are

usually wide than column headings but should be kept as narrow as possible without sacrificing precision and clarity of statements.

5. Body The body of the table contains the numerical information. This is the most vital part of the table. Data presented in the body arranged according to description are classifications of the captions and stubs.

6. Headnote It is a brief explanatory statement applying to all or a major part of the material in the table, and is placed below the point centered and enclosed in brackets. It is used to explain certain points relating to the whole table that have not been included in the title nor in the captions or stubs. For example, the unit of measurement is frequently written as a headnote, such as "in thousands" or "in million tonnes" or "in crores", etc.

7. Footnotes Anything in a table which the reader may find difficult to understand from the title, captions and stubs should be explained in footnotes. If footnotes are needed they are placed directly below the body of the table. Footnotes are used for the following main purposes:

(a) To point out any exceptions as to the basis of arriving at the data. for example, sales recorded at 'ex-factory price' for some of the entries and at 'delivered price' for others. Any heterogeneity in the data recorded must be disclosed to avoid wrong conclusions.

(b) Any special circumstances affecting the data, for example, strike, lock-out, fire, etc.

(c) To clarify anything in the table.

(d) To give the source in case of secondary data. The reference to the source should be complete in itself. For example, if the data is obtained from some periodical, its name, date of publication, page number, table number, etc., should be mentioned so that if the user wishes to check the data from the original source or considers later data from the same source he will know where to look for the information.

There are various systems of identifying the footnotes. One is numbering them consecutively with small numbers 1, 2, 3, 4 or letters a, b, c, d. Another system identifies the first footnote with one star (*), second footnote with two stars (**), third footnote with three stars (***), and so on. Sometimes instead of stars another sign † (dagger) is used. However, where several footnotes are required it is more convenient to use small letters.

The following is a specimen of table indicating the above parts:

Format of a Table

Title	*Headnote*	
Stub Heading	*Caption heading—Column heading*	
Stub Entries	Body	
Footnotes		Table Number

INTRODUCTION

In the previous chapter we have discussed the techniques of classification and tabulation that help in summarising the collected data and presenting them in a systematic manner. However, these forms of presentation do not always prove to be interesting to the common man. Too many figures are often confusing and may fail to convey the message effectively to those for whom they are meant.

One of the most convincing and appealing ways in which statistical results may be presented is through diagrams and graphs. Evidence of this can be found in newspapers, magazines, journals, advertisements, etc. There are numerous ways in which statistical data may be displayed pictorially such as different types of diagrams, graphs and maps. Very often the problem is that of selecting the best out of several methods that may be available. This is a difficult task and requires a great deal of artistic talent and imagination on the part of the individual or agency engaged in the preparation of diagrams and graphs. It is not practicable to discuss all the possible forms of charts here. An attempt is made in this chapter to illustrate some of the major types of diagrams, graphs and maps frequently used in presenting statistical data.

SIGNIFICANCE OF DIAGRAMS AND GRAPHS

Diagrams and graphs are extremely useful because of the following reasons:

They give a bird's-eye view of the entire data and, therefore, the information presented is easily understood. It is a fact that as the number and magnitude of figures increases they become more confusing and their analysis tends to be more strenuous. Pictorial presentation helps in proper understanding of the data as it gives an interesting form to it. The old saying 'A picture is worth 10,000 words' is very true. The mind through the eye can more readily appreciate the significance of figures in the form of pictures than it can follow the figures themselves.

They are attractive to the eye. Figures are dry but diagrams delight the eye. For this reason diagrams create greater interest than cold figures. Thus, while going though journals and newspapers the readers generally skip over the figures but most of them do look at the diagrams and graphs. Since diagrams have attraction value, they are very popular in exhibitions, fairs, conferences, board meetings and public functions.

They have a great memorising effect. The impressions created by diagrams last much longer than those created by the figures presented in a tabular form.

They facilitate comparison of data relating to different periods of time of different regions. Diagrams help one in making quick and accurate comparison of data. They bring out hidden facts and relationship and can stimulate as well as aid analytical thinking and investigation.

Comparison of Tabular and Diagrammatic Presentation

Data may be presented in the form of tables as well as diagrams and graphs. Both forms of presentation have their own usefulness for particular purposes. Hence, the choice of the form of presentation must be made

TYPES OF DIAGRAMS

In practice, a very large variety of diagrams are in use and new ones a constantly being added. It would be outside the scope of this book to de exhaustively with the subject and as such only the more frequently use diagrams are disussed. For the sake of convenience and simplicity the may be divided under the following heads:

One-dimensional diagrams, *e.g.*, bar diagrams.
Two-dimensional diagrams, *e.g.*, rectangles, squares and circles.
Three-dimensional diagrams, *e.g.*, cubes, cylinders and spheres.
Pictograms and cartograms.

Each of these types is discussed in detail in the following pages.

One-dimensional or Bar Diagrams

Bar diagrams are the most common type of diagrams used in practice. A bar is a thick line whose width is shown merely for attention. They are called one-dimensional because it is only the length of the bar that matters and not the width. When the number of items is large, lines may be

awn instead of bars to economise space. The special merits of bar agrams are the following:

(i) They are readily understood even by those unaccustomed to reading charts or those who are not chart-minded.
(ii) They possess the outstanding advantage that they are the simplest and the easiest to make.
(iii) When a large number of items are to be compared they are the only form that can be used effectively.

While constructing bar diagrams, the following points should be kept in mind:

(i) The width of the bars should be uniform throughout the diagram.
(ii) The gap between one bar and another should be uniform throughout.
(iii) Bars may be either horizontal or vertical. The vertical bars should be preferred because they give a better look and also facilitate comparison.
(iv) While constructing the bar diagram, it is desirable to write the respective figures at the end of each bar so that the reader can know the precise value without looking at the scale. This is particularly so where the scale is too narrow, for example, 1" on paper may represent 10 crore people. The two diagrams below would clarify the difference.

The fixed assets of Oswal Fats and Oils Ltd. from 2000-01 to 2002-03 re given below:

Year	:	*2000-01*	*2001-02*	*2002-03*
Fixed Assets (Rs. lakhs)	:	415	481	607

Represent the above data by a bar diagram.

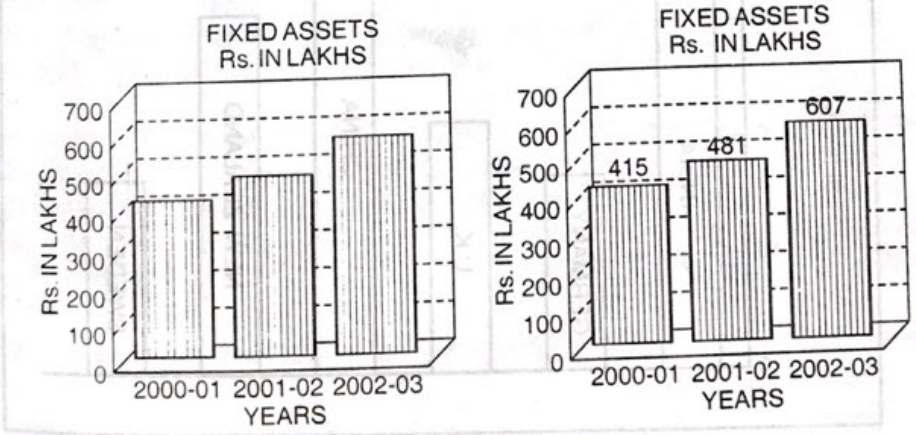


Fig.

From the second diagram one can easily make out the exact sales but rom the first it is difficult.

Types of Bar Diagrams

Bar diagrams are of the following types:

(a) Simple bar diagrams
(b) Sub-divided bar diagrams
(c) Multiple bar diagrams
(d) Percentage bar diagrams
(e) Deviation bars.

a) Simple Bar Diagrams A simple bar diagram is used to represent only one variable. For example, the figures of sales, production, population, etc., for various years may be shown by means of a simple bar diagram. Since these are of the same width and only the length varies, it becomes very easy for the reader to study the relationship. Simple bar diagrams are very popular in practice. This can be either vertical or horizontal. In practice, vertical bars are more popular. However, an important limitation of such diagrams is that they can present only one classification or one category of data. For example, while presenting the population for the last five decades, one can only depict the total population in the simple bar diagrams, and not its sex-wise distribution.

Illustration 1. Following table gives the birth rate per thousand of different countries over a certain period :

Country	*Birth Rate*	*Country*	*Birth Rate*
India	33	China	40
Germany	16	New Zealand	30
U.K.	20	Sweden	15

Represent the above data by a suitable diagram.

Solution. The above data can suitably is represented by simple bar diagram.

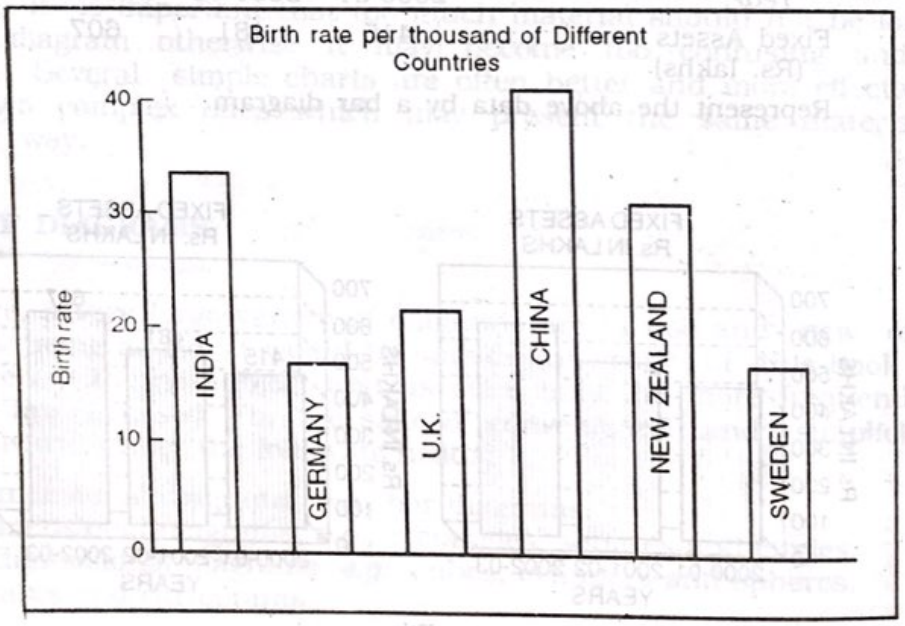


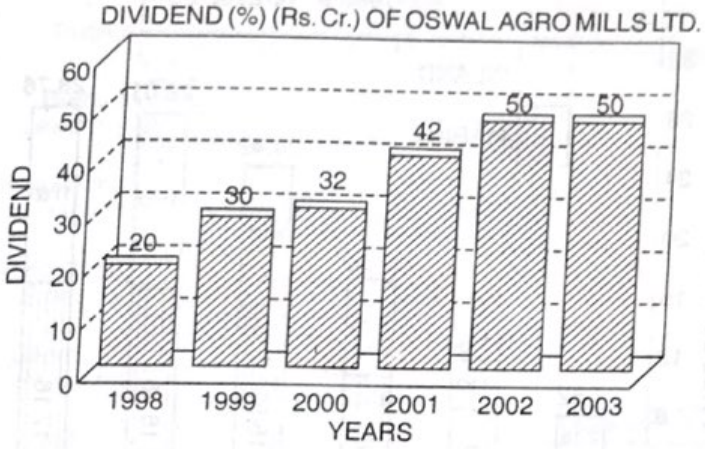
Illustration 2. The dividend given by Oswal Agro Mills Ltd. from 1998 to 2003 is given below :

Year (Rs. Crore) :	1998	1999	2000	2001	2002	2003
Dividend (%) :	20	30	32	42	50	50

Represent the data by a bar diagram.

* Such diagrams are also known as component bar diagrams.

Solution. The above data can be represented by a simple bar diagram:



(b) Sub-divided Bar Diagrams In a sub-divided bar diagram each bar representing the magnitude of a given phenomenon is further sub-divided in its various components. Each component occupies a part of the bar proportional to its share in the total.

For example, the number of students in various courses for B.Com., M.Com., B.A. and M.A. in various colleges may be represented by a sub-divided bar diagram. While constructing such a diagram, the various components in each bar should be kept in the same order. A common and helpful arrangement is that of presenting each bar in the order of magnitude from the largest component at the base of the bar to the smallest at the end. To distinguish between the different components, it is useful to use different shades or colours. Index or key should be given explaining these differences.

Sub-divided bar diagrams should not be used where the number of components is more than 10 or 12, for, in that case, the diagram will be overloaded with information which cannot be easily compared and understood.

The component bar diagram can be used to represent either the absolute data or distribution ratios such as percentage distributions. It is, in fact, an excellent method for presenting a set of distribution ratios diagrammatically.* The sub-divided bar diagrams can be constructed both on horizontal and vertical bases.

Illustration 3. The growth of production of fish for the period 1996-97 to 2002-03 is given below :

(lakh tonnes)

Year	*Marine*	*Inland*	*Total*
1996-97	5.34	2.18	7.52
1997-98	8.80	2.80	11.60
1998-99	10.86	6.70	17.56
1999-00	15.55	8.87	24.42
2000-01	16.98	11.03	28.01
2001-02	17.16	11.60	28.76
2002-03	12.47	8.42	20.89

Represent the data by a suitable diagram.

* The other alternatives for this purpose are the relative pie diagram and the relative component line chart. The latter can be used only in cases where the classification is chronological. When the number of time periods is not large, t

Solution. The above data can be represented by a sub-divided bar diagram (drawn o vertical base,

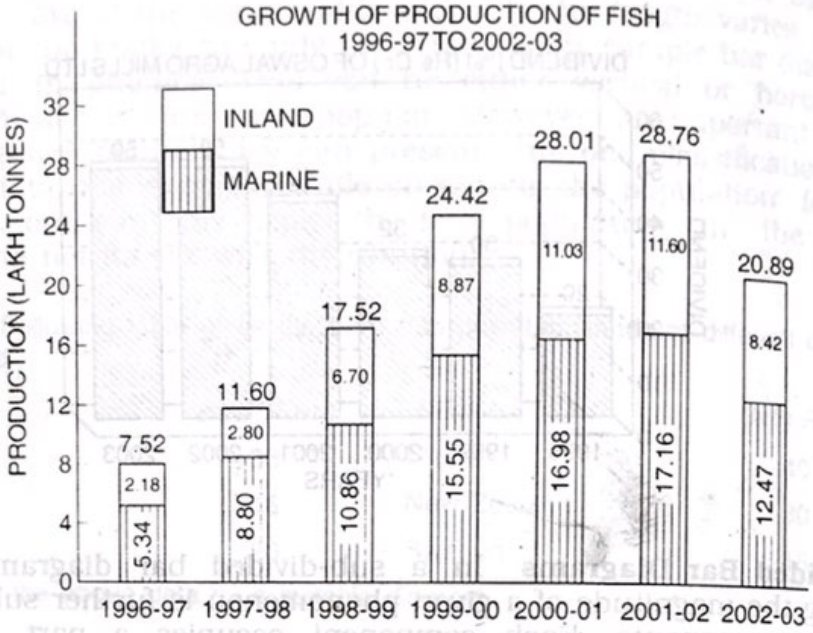


Illustration 4. The following data relate to exports in 2001-02 (provisional) and expo target for 2002-03:

Figure (Rs. crore)

	2001-02	*2002-03*
Textiles	15185	23680
Gems and jewellery	8839	11056
Engineering goods	6505	8000
Agricultural products	5514	7216
Chemical products	5419	6938
Leather products	3692	4768
Ores and minerals	2146	2781
Marine products	1743	2080
Plantations	1339	1600
Electronics	611	1200
Petroleum products	1379	861
Raw cotton	196	544

Represent the data by horizontal bars. (Please see page 135 for diagram)

(c) Multiple Bars In a multiple bar diagram two or more sets of interrelated data are represented. The technique of drawing such a diagram is the same as that of simple bar diagram. The only difference is that since more than one phenomenon is represented, different shades, colours, dots or crosses are used to distinguish between the bars. Wherever a comparison between two or more related variables is to be made, multiple bar diagram should be preferred.

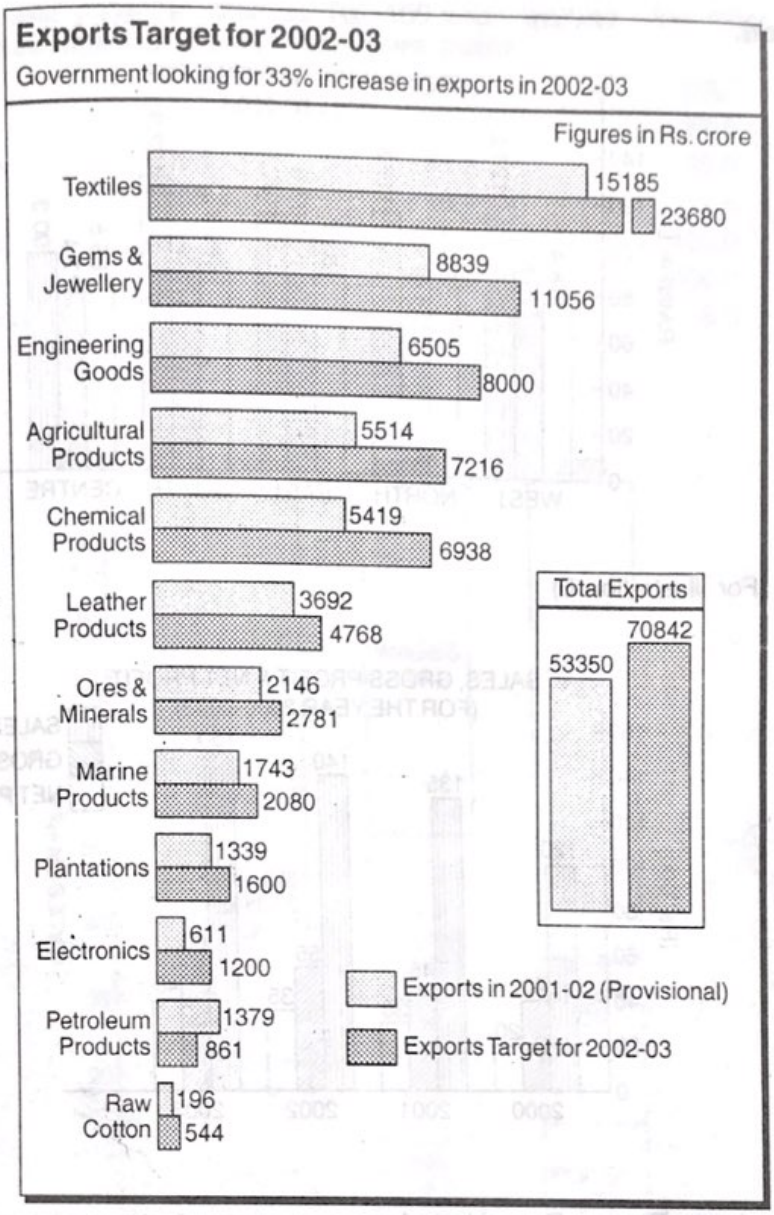


Illustration 5. The regional rainfall indices during the year 2001 to 2003 are given below:

Year	Zone				
	West	*North*	*East*	*South*	*Centre*
2001	78.4	88.9	83.7	89.9	86.5
2002	75.6	62.5	103.6	75.5	77.4
2003	121.2	116.5	107.6	123.9	90.3

Represent the data by a multiple bar diagram. (Please see page 136 for diagram)

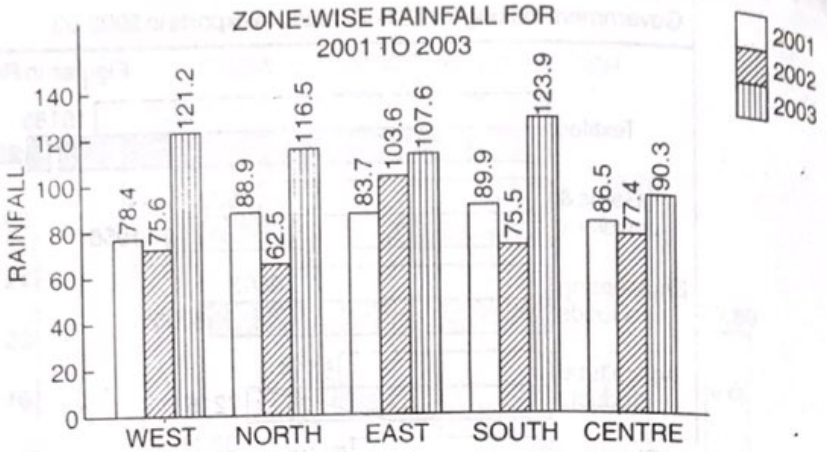
Illustration 6. Draw a multiple bar diagram from the following data:

Year	*Sales ('000 Rs.)*	*Gross Profit ('000 Rs.)*	*Net Profit ('000 Rs.)*
2000	120	40	20
2001	135	45	30
2002	140	55	35
2003	150	60	40

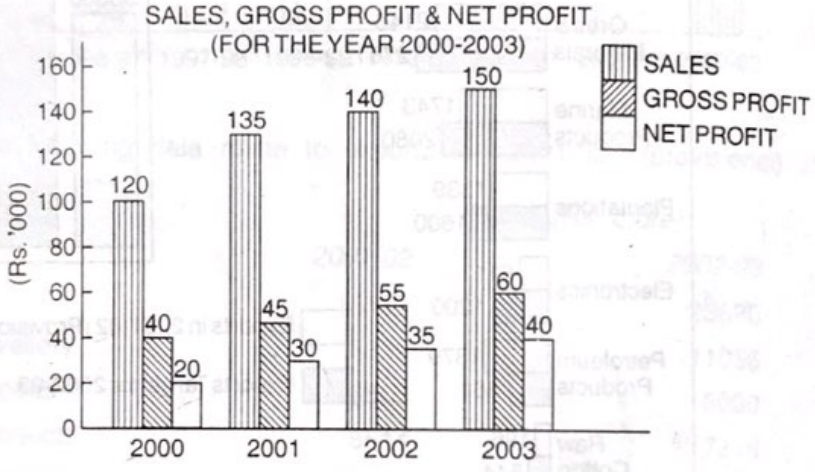
(B.Com., S.V. Univ.; B.Com., Gulbarga Univ.)

(Please see page 136 for diagram)

Solution.



Solution. (For illustration 6)



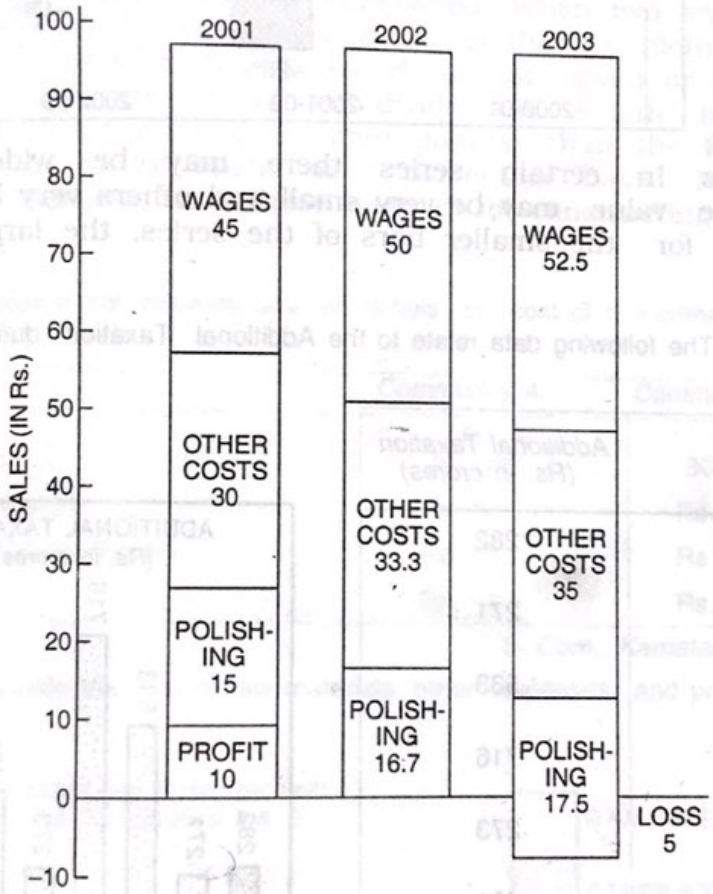
(d) Percentage Bars Percentage bars are particularly useful in statistical wo which requires the portrayal of relative changes in data. When such diagram are prepared, the length of the bars is kept equal to 100 and segments are c in these bars to represent the components (percentages) of an aggregate.

Illustration 7. Represent the following by sub-divided bar diagram drawn on the pe centage basis:

Particulars	*2001*	*2002*	*2003*
1. Cost per chair			
(a) Wages	9	15	21
(b) Other costs	6	10	14
(c) Polishing	3	5	7
Total	18	30	42
2. Proceeds per chair	20	30	40
3. Profit (+)	+2	–	–2
Loss (–)			

Solution. Take the sale price per chair as Rs. 100 and express the other figures in percentages. The percentages so obtained are given below:

Particulars	*2001*	*2002*	*2003*
Wages	45.0	50.0	52.5
Other costs	30.0	33.3	35.0
Polishing	15.0	16.7	17.5
Total costs	90.0	100.0	105.0
Sale price	100.0	100.0	100.0
Profit or loss	+10.0	—	−5.0



(e) Deviation Bars Deviation bars are popularly used for representing net quantities—excess or deficit, *i.e.*, net profit, net loss, net exports or net imports, etc. Such bars can have both positive and negative values. Positive values are shown above the base line and negative values below it. The following illustration would explain this type of diagram:

Illustration 8. Present the following data by a suitable diagram showing the Sales and Net Profits of private industrial companies:

Year	*Sales*	*Net Profit or loss*
2000-01	14%	49%
2001-02	10%	−25%
2002-03	13%	−1%

Pie Diagram Pie diagrams are very popularly used in practice to show percentage breakdowns. For example, with the help of a pie diagram we can show how the expenditure of the Government is distributed over different heads like Agriculture, Irrigation, Industry, Transport, Defence, etc. Similarly through a pie diagram we can show how the expenditures incurred by an industry are divided under different heads like raw materials, wages and salaries, selling and distribution expenses, etc. The pie chart is so called because the entire graph looks like a pie, and the components resemble slices cut from pie.

While making comparisons, pie diagrams should be used on a percentage basis and not on an absolute basis, since a series of pie diagrams showing absolute figures would require that larger totals be represented by larger circles. Such presentation involves difficulties of two-dimensional comparisons. However, when pie diagrams are constructed on a percentage basis, percentages can be presented by circles equal in size. It may be noted that this problem does not arise in the use of a single pie diagram.

In laying out the sectors for pie chart, it is desirable to follow some logical arrangement or sequence. It is common practice to begin the largest component sector of pie diagram at 12 O'clock position on the circle. Usually the other component sectors are placed in clockwise succession in descending order of magnitude, except for catch-all components like "miscellaneous" and "all others" which are shown last, contrast with adjacent sectors.

In constructing a pie chart the first step is to prepare the data so that the various component values can be transposed into corresponding degrees on the circle. Suppose there are four components in a series representing the following values : *(i)* 60 per cent, *(ii)* 25 per cent, *(iii)* (360/100 = 3.6), the corresponding values of the four components in the illustration are (60) × (3.6) = 216'; (25) × (3.6) = 90'; (10) × (3.6) = 36'; (5) × (3.6) = 18'.

The second step is to draw a circle of appropriate size with a compass. The size of the radius depends upon the available space and other factors of presentation.

The third step is to measure points on the circle representing the size of each sector with the help of protractor. The ordinary protractor is based upon a scale in which the total circle is 360 degrees, but it is possible to purchase a protractor in which the entire circle is divided not into 360 but 100 equal parts so that the angle representing any desired percentage can be read directly.

In laying out the sectors for a pie chart it is desirable to follow some logical arrangement, pattern or sequence. For example, it is a common procedure to arrange the sectors according to size, with the largest at the top and others in sequence running clockwise. An essential feature of pie chart is the careful identification of each sector with some kind of explanatory or descriptive label. If there is sufficient room, the labels can be placed inside the sectors; otherwise the labels should be placed in contiguous positions outside the circle, usually with an arrow pointing towards the appropriate sector.

* (pie) 22. We have divided the outlays by h to get the square of radius.

Illustration 14. Draw a pie diagram for the following data of Sixth Five-Year Plan Pub Sector outlays:

Agriculture and Rural Development	12.9%
Irrigation, etc.	12.5%
Energy	27.2%
Industry and Minerals	15.4%
Transport, Communication, etc.	15.9%
Social Services and Others	16.1%

(B.Com., Bangalore Univ., 199

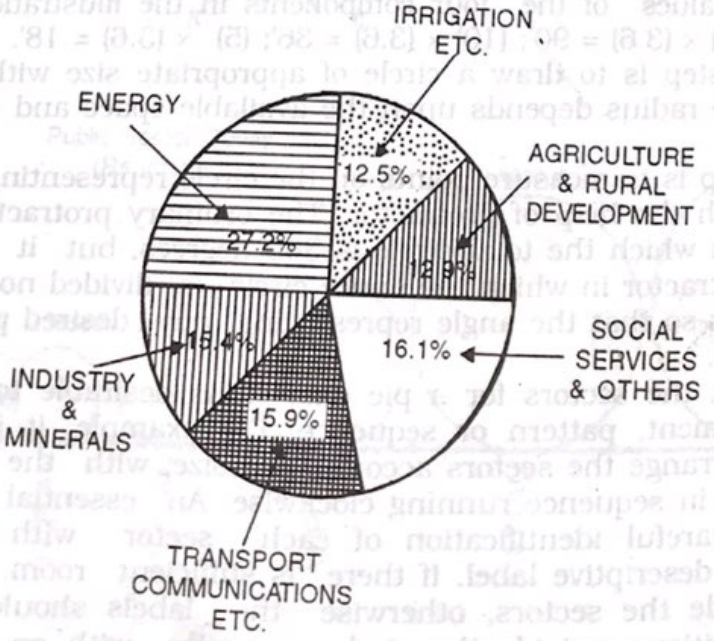
Solution. The angle at the centre is given by

$$\frac{\text{Percentage outlay}}{100} \times 360 = \text{Percentage outlay} \times 3.6$$

COMPUTATION FOR PIE DIAGRAm

Sector	*Percentage*	*Angle outlay*
Agriculture and Rural Development	12.9	$12.9 \times 3.6 = 46°$
Irrigation, etc.	12.5	$12.5 \times 3.6 = 45°$
Energy	27.2	$27.2 \times 3.6 = 98°$
Industry and Minerals	15.4	$15.4 \times 3.6 = 56°$
Transport, Communication, etc.	15.9	$15.9 \times 3.6 = 57°$
Social Services and others	16.1	$16.1 \times 3.6 = 58°$
Total	100.0	360°

Now a circle shall be drawn suited to the size of the paper and divided into 6 parts according to degrees of angle at the centre. (The angles have been arranged in descending order).



satisfactorily by some other means, such as conventional percentage

GRAPHS OF FREQUENCY DISTRIBUTIONS

A frequency distribution can be presented graphically in any of the following ways:

- Histogram,
- Frequency polygon,
- Smoothed frequency curve, and
- 'Ogives' or cumulative frequency curves.

Histogram*

Out of several methods of presenting a frequency distribution graphically, histogram is the most popular and widely used in practice. A histogram is a set of vertical bars whose areas are proportional to the frequencies represented.

While constructing histogram the variable is always taken on the *X*-axis and the frequencies depending on it on the *Y*-axis. Each class is then represented by a distance on the scale that is proportional to its class-interval. The distance for each rectangle on the *X*-axis shall remain the same in case the class-intervals are uniform throughout. If they are different they vary. The *Y*-axis represents the frequencies of each class which constitute the height of its rectangle. In this manner we get a series of rectangles each having a class-interval distance as its width and the frequency distance as its height. The area of the histogram represents the total frequency as distributed throughout the classes.

The histogram should be clearly distinguished from a bar diagram. The distinction lies in the fact that where a bar diagram is one dimensional, *i.e.*, only the length of the bar is material and not the width, a histogram is two-dimensional, that is, in a histogram both the length as well as the width are important.

The histogram is most widely used for graphical presentation of a frequency distribution. However, we cannot construct a histogram for distribution with open-end classes. Moreover, a histogram can be quite misleading if the distribution has unequal class-intervals and suitable adjustments in frequencies are not made.

The technique of contracting histogram is given below *(i)* for distributions have equal class-intervals, and *(ii)* for distributions having unequal class-intervals.

When class-intervals are equal, take frequency on the *Y*-axis, the variable on the *X*-axis and construct adjacent rectangles. In such a case the height of the rectangles will be proportional to the frequencies.

When class-intervals are unequal, a correction for unequal class intervals must be made. The correction consists of finding for each class the frequency density or the relative frequency density. The frequency density is the frequency for that class divided by the width of that class. A histogram or frequency density polygon constructed from these density values would have the same general appearance as the corresponding graphical display developed from equal class intervals.

For making the adjustment we take that class which has lowest class-interval and adjust the frequencies of other classes in the following manner. If one class-interval is twice as wide as the one having lowest class-interval we divide the height of its rectangle by two, if it is three times more we divide the height of its rectangle by three, etc., *i.e.*, the heights will be proportional to the ratio of the frequencies of the width of the class.

The following illustration would clearly indicate the manner in which the adjustment has to be made:

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* A histogram is a bar chart or graph showing the frequency of occurrence of each value of the variable being analysed. —*Opermann*

Illustration 25. Represent the following data by a histogram:

Marks	No. of Students	Marks	No. of Students
0–10	8	50–60	60
10–20	12	60–70	52
20–30	22	70–80	40
30–40	35	80–90	30
40–50	40	90–100	5

Solution. Since class-intervals are equal throughout no adjustment in frequencies is required.

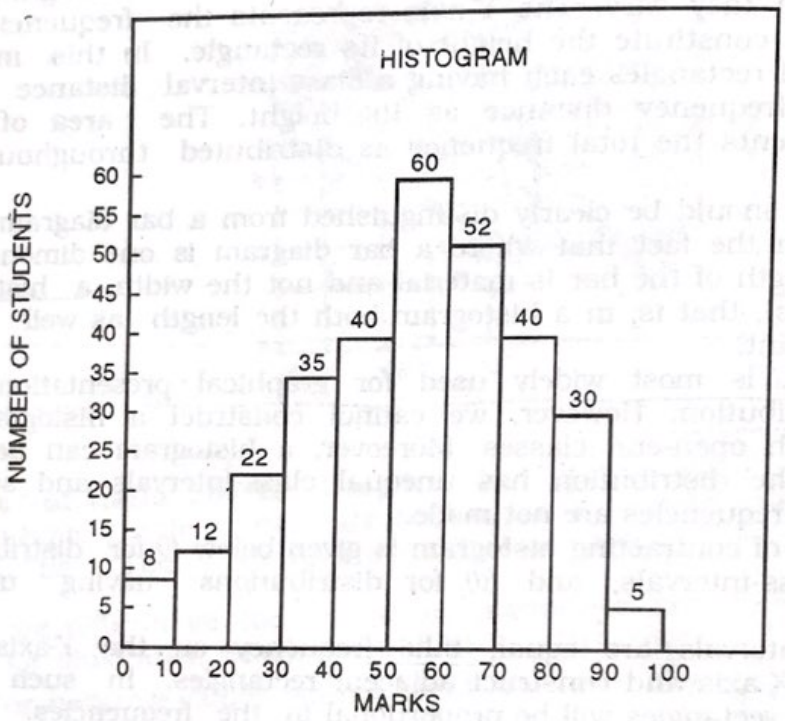


Illustration 26. Represent the following data by means of a histogram:

Weekly Wages (in Rs.)	No. of Workers	Weekly Wages (in Rs.)	No. of Workers
10–15	7	30–40	12
15–20	19	40–60	12
20–25	27	60–80	8
25–30	15		

(B. Com. (H)., Delhi Univ., B. Com., A.P. Univ.)

Solution. Since the class-intervals are unequal, frequencies must be adjusted otherwise the histogram would give a misleading picture. The adjustment is done as follows. The lowest class interval is 5. The frequencies of the class 30-40 shall be divided by two since the class interval is double, that of 40-60 by 4, etc.

Construction of Histogram when only Mid-points are given When only mid-points are given, ascertain the upper and lower limits of the various classes and then construct the histogram in the same manner.

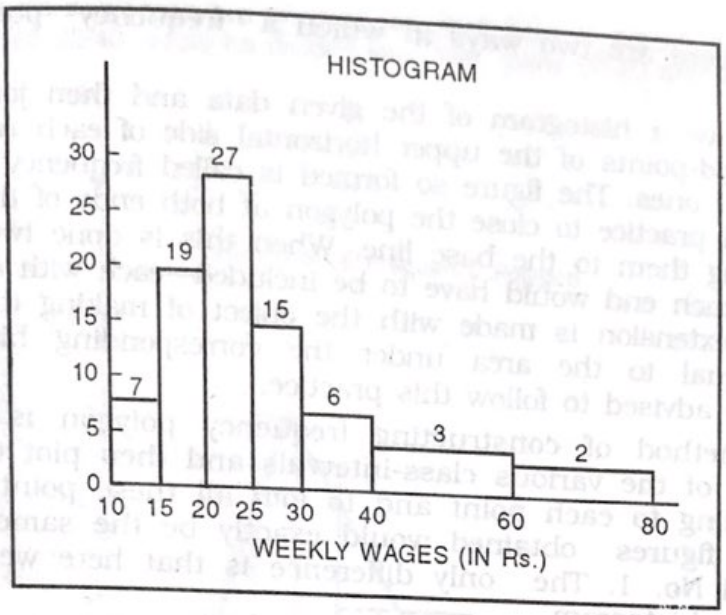
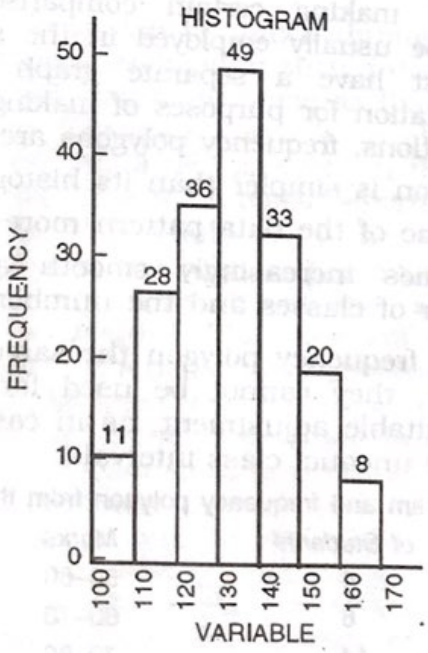


Illustration 27. Draw the histogram for the following data:

Variable	*Frequency*	*Variable*	*Frequency*
100–110	11	140–150	33
110–120	28	150–160	20
120–130	36	160-170	8
130–140	49		

(B.A., Coimbatore Univ., 1997)



Frequency Polygon*

A frequency polygon is a graph of frequency distribution. It has more than four sides. It is particularly effective in comparing two or more frequency

* "Polygon" literally means 'many-angled' diagram.